Drivers of sustainability: technological and relational factors influencing young consumers' green buying intentions and green actual consumption behavior

Naseer Abbas Khan, Sajid Hassan, Natalya Pravdina and Maria Akhtar

Naseer Abbas Khan is based at the Department of Industrial Economics and Project Management, South Ural State University, Chelyabinsk, Russian Federation. Sajid Hassan is based at the Department of Psychology, International Islamic University, Islamabad, Pakistan. Natalya Pravdina is based at the Department of Industrial Economics and Project Management, South Ural State University, Chelyabinsk, Russian Federation. Maria Akhtar is based at Malik Firoz Khan Noon Business School, University of Sargodha, Sargodha, Pakistan.

Abstract

Purpose – *This study aims to explore the factors that influence young green actual consumption behavior (GACB) by examining the relational and technological aspects that are dependent on technology and youth green buying intention (GBI). Additionally, this study endeavors to examine the moderating effect of adolescent green organizational citizenship behavior (OCB) on the association between young consumer GBI and young GACB.*

Design/methodology/approach – *The data for this study was collected from a sample of 282 university students located in the southern region of China. A time lag approach was used in this study, with data being gathered during two separate intervals spaced apart by duration of two months.*

Findings – *The findings of this study demonstrate that both technological and relational factors play a significant role in predicting young consumer GBI, which in turn predicts young GACB. Additionally, the results indicate that GBI is a crucial facilitator in the investigation. Furthermore, the results reveal that young green OCB serves as a significant moderator, enhancing the association between young consumer GBI and young consumer GACB.*

Originality/value – *The present study provides a novel perspective on the examination of how technology and relational factors impact young consumers, offering a deeper understanding of their intentions and actual consumption behavior. The findings of this study offer both theoretical and practical implications for academics, policymakers, senior managers and practitioners, providing valuable insights into the field.*

Keywords *Technological factor, Relational factors, Green buying intention, Green consumption behavior, Green organizational citizenship behavior, Young consumer*

Paper type *Research paper*

1. Introduction

The advent of digital and online shopping has presented both opportunities and challenges for consumers (Ali *et al.*, 2022). In terms of opportunities, digital and online shopping provides consumers with convenience, accessibility and a wider range of products (Khan *et al.*, 2020a, 2020b). Consumers can easily purchase products from the comfort of their homes, which saves time and eliminates the need to physically visit stores (Huang *et al.*, 2021). Furthermore, online shopping enables consumers to compare prices, read reviews and make informed purchasing decisions (Khan *et al.*, 2023). However, online shopping also presents challenges for consumers. One of the challenges is the risk of fraud and identity theft (White and Fisher, 2008). Consumers prefer to exercise caution when disclosing personal information online, as cybercriminals can use it to steal their identities and money (Wang *et al.*, 2021). Additionally, online shopping can lead to overspending and

Received 27 September 2022
Revised 14 February 2023
30 March 2023
15 May 2023
Accepted 17 June 2023

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

impulse buying, as consumers can easily make purchases with just a few clicks (Zhao *et al.*, 2021). A further obstacle related to customers is their inclination to make environmentally conscious choices while engaging in online shopping (Guath *et al.*, 2022). As consumers' desire to purchase environmentally sustainable products persists in the digital era, it can lead to heightened awareness of the impact of consumerism on the environment (Barbarossa and Pastore, 2015).

In this regard, studies have shown that several factors can influence consumers' green consumption behavior, including personal values, attitudes, perceived benefits and perceived behavioral control (Ajzen, 1991; Khan and Khan, 2021). Technological factors, such as the perceived ease of use and perceived usefulness of digital and online shopping platforms, can also play a significant role in shaping young consumers' green consumption behavior (Venkatesh and Davis, 2000). Gao *et al.* (2022) found that online platforms on social media effectively attract users, facilitate socialization and engagement processes and provide them with appropriate information to make well-informed decisions. The integration of social media and information systems has been found to offer valuable data that can aid organizations in making informed decisions (Khan *et al.*, 2022).

Despite previous research examining the use of technology in shaping consumer buying behavior, few studies have investigated the impact of technological on consumers' eco-friendly behavior (Wang, 2019; Zhang *et al.*, 2018). Furthermore, prior research has focused on technological factors for consumer trust, intention and behavior in traditional shopping (Fan *et al.*, 2020; Faqih, 2016), rather than green intention and behavior in online shopping. There is a gap in the literature that requires further investigation to gain a deeper understanding of the factors influencing young consumers' green intentions and behavior in online shopping. Scholars are constantly striving to identify gray or uncovered areas in academic literature, and this study strives to address the scholarly gap by exploring the interplay between technological and relational factors on young consumers' green intentions and behavior.

Previous research has examined technological and relational factors separately, there is a need to investigate the impact of their interaction on young consumers' behavioral outcomes. For example, previous research has shown that the interaction between website design and social norms affects young consumers' green purchase intentions (Ahmed *et al.*, 2021), while the interaction between trust in online retailers and the availability of eco-friendly product information on the website influences young consumers' green purchase intentions (Ng and Cheung, 2022). Based on insufficient empirical findings of the past studies encourage us to investigate the interplaying role of mediator (i.e. green buying intention [GBI]) and moderator (environmental attitude) in framing our specific research objectives.

The main objective of this study is to assess how technological and relational factors affect young consumers' GBI and explore the subsequent impact of GBI on their green actual consumption behavior (GACB). There is a need to investigate the role of mediator, specifically GBI, and moderator, namely, green organizational citizenship behavior (OCB), in shaping our specific research objectives. This will enable a deeper understanding of the relationship between these factors and how they influence consumer behavior. This study has two primary objectives: first, to investigate the mediating role of GBI in the relationship between technological and relational factors and green attitude toward eco-friendly consumer behavior (GACB). Empirical evidences indicate the significance of underlying mechanism such as intention (Turchi and Steg, 2009). Second, to evaluate the potential moderating effect of green OCB on the relationship between GBI and GACB. By achieving these objectives, this study aims to provide a comprehensive understanding of the factors that influence consumer behavior toward eco-friendly products, which can help promote sustainable consumption practices. As a moderating variable, OCB measures how much a person practices environmentally beneficial habits outside of the scope of their official job

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

duties (Organ, 1988). OCB has been shown to enhance the association between consumer intention and buying behavior (Khan and Khan, 2021).

This study may provide valuable insights into the role that GBI plays in shaping young consumer GACB by examining the moderating effect of green organizational OCB. Understanding the importance of OCB in adolescent GACB will also be important for policymakers and organizations that aim to promote sustainable consumption behaviors among young people. Moreover, theory of planned behavior (TPB) is a well-established framework that explains how attitudes, subjective norms and perceived behavioral control can influence behavioral intentions and actual behavior (Ajzen, 1985). The TPB can be used to explain the relationship between young consumer GBI and young GACB. Additionally, it can also help in examining how the relational and technological aspects that are dependent on technology affect GBI and GACB The importance of trust and rapport in shaping consumer behavior has been well established in the literature (Anderson and Narus, 2004), and these factors are likely to play a key role in shaping young consumers' intentions and behaviors related to green consumption.

2. Theory and hypothesis

2.1 Theoretical support

TPB is a social psychology theory used to explain and predict human behavior. According to the TPB, human behavior is influenced by three factors: attitudes, subjective norms and perceived behavioral control (Ajzen, 1985). TPB suggests that behavioral intentions and subsequent actions can be influenced by attitudes, subjective norms and perceived behavioral control. We can better understand how technology aspects such as perceived usefulness and ease of use affect GBI and purchasing decisions by using the TPB to analyze consumer GBI. Perceived usefulness reflects the degree to which a technology can aid in achieving a specific goal or objective, whereas perceived ease of use reflects the perceived level of difficulty associated with using a particular technology. These factors are significant in determining consumers' intentions to adopt behaviors (Rana and Paul, 2017) and purchase environmentally friendly products in the context of green behavior.

Several studies have applied the TPB to understand the relationship between technology and behavioral outcome. Qi *et al.* (2021) found that perceived ease of use and perceived usefulness significantly influenced consumers' attitudes toward green products. Similarly, Yuan *et al.* (2016) found that perceived ease of use and perceived usefulness were significant predictors of consumers' green behavior intentions, which in turn influenced their purchase behavior. Likewise, TPB can be considered as a viable and rational approach to address the influence of relational factors such as "rapport and trust", as these factors stem from the connection between the workforce and the organization, facilitating the formation of attitudes and actions that foster a sustainable organization. Rapport refers to the degree to which a consumer feels a sense of connection and similarity with a company or organization (Gremler and Gwinner, 2000), while trust refers to the degree to which a consumer perceives a company or organization as reliable and honest (Chen and Dhillon, 2003). In the context of green behavior, these factors can impact consumers' intentions to engage in environmentally friendly behaviors, engage in green OCB and purchase green products.

Past studies have applied the TPB to explain impact of relational factors on employees behavioral outcome For example, Hollebeek and Macky (2019) found that consumers' perceptions of trust in a company were positively related to extra-role behavior. In this context, consumers' perceptions of rapport with a company were positively related to their attitudes toward green products (Kim and Yun, 2019). Furthermore, Qi and Ploeger (2019) found that trust in a company was a significant predictor of consumers' intentions to engage in green behavior and purchase green products, while subjective norms and perceived behavioral control were also important factors.

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

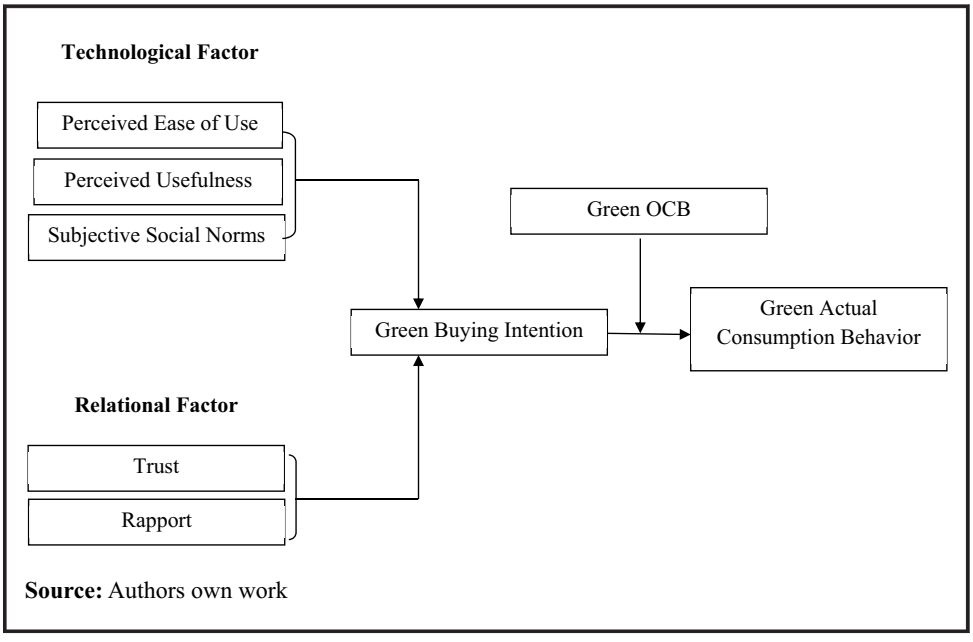
To summarize, our proposed research model depicted in Figure 1 offers a viable means of expanding the applicability of TPB, serving as a valuable framework to comprehend the effects of technological and relational factors on consumers' green behavioral intentions, green OCB and purchase behavior. TPB can assist in finding key drivers that influence customers' decisions to engage in eco-friendly behaviors, display green OCB and purchase ecologically sustainable items by taking into account consumers' attitudes, subjective norms and perceived behavioral control.

2.2 Hypothesis development

The primary objective of this study is to ascertain the impact of both relational and technological factors on GACB, using consumer GBI as a mediator and green OCB as a moderator. It is noteworthy that TPB places substantial emphasis on the concept of behavioral intention, which is considered the most direct predictor of human social behavior (Ajzen, 2005). By recognizing the critical role of behavioral intention, TPB offers an important framework for predicting and understanding human behavior (Ajzen, 1991, 2011). This theory has been used as a theoretical framework in the current study to analyze the factors that influence young GACB by investigating the relational and technological factors that are dependent on technology and GBI. According to Ajzen (1991), TPB maintains that behavioral intent is the most accurate predictor of human social behavior. Attitudes, subjective standards and perceived behavioral control are the three major tenets of the theory, which all have an effect on intention.

Perceived ease of use can be linked to the level of expected performance, whereas perceived usefulness can be linked to the degree of expected effort required for technology adoption and usage. Subjective social norms are a part of the TPB framework and refer to the perceived social pressure or influence from the beliefs and attitudes of significant others toward a specific behavior (Davis, 1989). In other words, subjective social norms represent the influence that other people have on an individual's attitudes and behaviors. Studies have shown that higher levels of perceived ease of use (Lin and Lu, 2011; Wang *et al.*, 2013) and

Figure 1 Research model



Source: Authors own work

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

perceived usefulness (Chen and Chang, 2014) increase the likelihood of adopting green technology.

Likewise, subjective social norms have been identified as a critical determinant in influencing the ecologically conscious behavior of young consumers (Chen and Chang, 2014). Furthermore, studies have shown that perceived ease of use (Lin and Lu, 2011; Wang *et al.*, 2013), and subjective social norms (Kim and Ko, 2010) are positively associated with young consumers' technology adoption behavior. The same argument can be applied to young consumers' green purchasing behavior as the use of technology can greatly impact and facilitate the process of making environmentally friendly choices (Lee *et al.*, 2012). Thus, it can be hypothesized as follows:

H1. Technological factors (perceived ease of use, perceived usefulness and subjective social norms) are positively associated with young consumer GBI.

Studies have shown that relational factors such as trust in the supplier of green technology (Chang, 2011; Wang and Kim, 2013) and rapport (Kim and Ko, 2010) play a crucial role in shaping consumers' buying intentions. Higher levels of trust increase the likelihood of adopting green technology (Chang, 2011) while a good rapport can increase the perception of the technology being useful and easier to use (Kim and Ko, 2010). In the context of green technology, social pressure and optimal conditions might be viewed as trust and rapport, respectively. These relationship factors have a big impact on how likely people are to purchase anything (Chang, 2011). Higher levels of trust in the supplier have been shown to increase the likelihood of adopting green technology (Chen and Chang, 2012), while a good rapport can increase the perception of the technology being useful and easier to use (Kim and Ko, 2010). The TPB provides a comprehensive framework for understanding the impact of relational factors on young people's green behavior by considering trust and rapport as key determinants of technology acceptance and use. Therefore, it can be hypothesized as follows:

H2. Relational factors (trust and rapport) are positively associated with young consumer GBI.

In the context of green technology, perceived ease of use, perceived usefulness and social norms have been identified as key determinants of technology acceptance and use (Davis, 1985; Kim and Ko, 2010). These factors have been shown to positively impact young consumer green intention (Chang, 2011). However, the direct association between these technological factors and GACB remains unclear. To address this gap in the literature, previous research has proposed the mediating role of green intention in the association between technological factors and green consumption behavior (Park *et al.*, 2014).

Studies have found that green intention serves as a mediating factor, linking technological factors and consumer consumption behavior (Chen and Chang, 2012; Park *et al.*, 2014). In other words, green intention acts as a bridge between the impact of perceived ease of use, perceived usefulness and social norms on GACB. For example, Kim *et al.* (2012) found that green intention significantly mediates the association between perceived ease of use and green consumption behavior, whereas Park *et al.* (2014) found that green intention significantly mediates the association between perceived usefulness and green behavior. The TPB offers a thorough framework for comprehending how technology factors affect young customers' purchasing decisions when they shop online. There is evidence to suggest that young people's beliefs about technology can have a significant impact on their technology adoption and use behaviors. For example, in a study by Kwon and Wen (2017), it was found that young people's attitudes toward technology were significant predictors of their technology adoption and use behaviors. Based on the arguments presented above, the following hypothesis can be proposed:

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

H3. Young consumer GBI has a significant mediating effect in the association between technological factors (perceived ease of use, perceived usefulness and subjective social norms) and young GACB.

Studies have found that green intention, or the intention to engage in environmentally responsible behavior, can play a crucial role in mediating the association between relational factors and actual consumption behavior. For example, Wang and Chen (2009) found that trust and rapport can positively impact a consumer's green intention, which in turn can impact their actual consumption behavior. Similarly, Wei and Lee (2014) found that green intention mediates the association between trust and actual consumption behavior, as consumers who trust and have a strong intention to engage in environmentally responsible behavior are more likely to make environmentally responsible purchases. The literature suggests that consumer green intention can play a significant mediating role in the association between relational factors (trust and rapport) and actual consumption behavior. Thus, it can be hypothesized as follows:

H4. Young consumer GBI has a significant mediating effect on the association between relational factors (trust and rapport) and young consumer GACB.

Several studies have investigated the association between green intention and green consumption behavior. These studies provide evidence supporting the proposed association between green intention and green consumption behavior among youth. For example, a study by Liang and Li (2017) found that green intention positively influenced green consumption behavior among youth. The authors found that environmental knowledge and subjective norms were important antecedents of green intention, while green consumption behavior was influenced by green intention and environmental attitudes. This study concluded that environmental education and social persuasion could be effective strategies to promote green consumption behavior among youth.

Another study by Kollmuss and Agyeman (2002) found that environmental attitudes and subjective norms were important predictors of green behavior among young adults. This study found that young adults who held positive environmental attitudes and felt pressure from others to adopt environmentally responsible behaviors were more likely to engage in green consumption behaviors. Lin and Hsiao (2020) investigated the association between green intention and green consumption behavior among college students in Taiwan. Green intention positively influenced green consumption behavior and that the association was mediated by environmental attitudes (Babalola *et al.*, 2020; Lin and Hsiao, 2020). Lin and Hsiao (2020) concluded that promoting positive environmental attitudes could be an effective strategy for encouraging green consumption behavior among college students. Past studies provided evidence supporting the proposed association between green intention and green consumption behavior (Chen and Chang, 2012; Kim and Lee, 2020). By highlighting the importance of environmental attitudes and subjective norms, these studies suggest that promoting positive environmental attitudes and social pressure could be effective strategies for promoting green consumption behavior among youth. Thus, we can hypothesize as follows:

H5. Young consumer GBI is positively associated with young consumer GACB.

Several studies have investigated the moderating role of green OCB. For example, Kim and Lee (2020) found that green OCB played a moderating role in the association between perceived usefulness and green intention in online shopping. The authors found that perceived usefulness positively influenced green intention, but this association was stronger for individuals who exhibited high levels of green OCB. Another study by Hu and Lu (2021) found that green OCB moderated the association between perceived ease of use and green intention in online shopping. The authors found that perceived ease of use positively influenced green intention, and this association was stronger for individuals who exhibited high levels of green OCB. Zhang and Wang (2022) found that green OCB

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

moderated the association between perceived risk and green intention in online shopping. The authors found that perceived risk negatively influenced green intention, but this association was weaker for individuals who exhibited high levels of green OCB.

Some other studies have investigated the moderating effect of green OCB on the association between green intention and green consumption behavior in online shopping, For instance, Xu and Zhang (2023) found that eco-friendly attitudes positively influenced green consumption behavior, and this association was stronger for individuals who exhibited high levels of green OCB. Liu and Chen (2022) found that environmental concern positively influenced green consumption behavior, and this association was stronger for individuals who exhibited high levels of green OCB. Thus, we can propose as:

H6. Young consumer green OCB significantly moderates the association between GBI and GACB in this way; as the young consumer green OCB increases, the association between GBI and GACB strengthens and vice versa.

3. Method

3.1 Data collection procedure

This study used survey to collect the data and surveys are commonly used in research because they allow researchers to collect large amounts of data from a large and diverse group of people in a relatively short amount of time. Surveys can be used to collect both quantitative and qualitative data and can be administered in a variety of formats, including online, paper, phone or in-person. Surveys can also be designed to measure a wide range of constructs, such as attitudes, opinions, behaviors, demographics and more. The data for this study was collected from university students in the Southern regions of China. Youth in university are engaged in online shopping and young consumers have become increasingly aware of environmental issues and are more likely to engage in pro-environmental behaviors, such as online green shopping (Liang and Li, 2017).

A questionnaire was used to collect data; it was initially developed in English and then translated into Chinese using translation-back translation technique suggested by Brislin (1980). A total of 553 university students studying in different universities in the Southern regions of China were targeted for the survey. Research sample was drawn on the basis of formula $n = [(Z\alpha/2 + Z\beta)/ES]$^2 Where n = sample size $Z\alpha/2$ = the critical value for the desired level of statistical significance, $Z\beta$ = the critical value for the desired power of this study, ES = the effect size of interest. This formula helps to draw sample size in this way $n = [(1.96 + 0.84)/0.50]$^2 $= 23.52$^2 ≈ 553.

The questionnaires were distributed during university hours, and participants were informed beforehand that their responses would be used strictly for academic purposes and that their confidentiality would be strictly maintained. In the first wave of data collection, 396 participants returned the completed questionnaires. In the second wave of data collection, the sample was narrowed down to 282 participants, which constituted the final sample for this study. Overall valid response rate was 51%. Data for this study was collected using a time-lag approach, where two different data collection points were separated by a time period of two months. The purpose of this time-lag approach was to provide participants with sufficient time to reflect and act before providing their responses. At Time 1, participants provided responses to demographic information, such as age, gender and educational background, as well as their technological factors, association factors and consumer GBI. At Time 2, the same participants were asked to provide their responses on GACB and green OCB. The demographic characteristics of the participants are presented in Table 1.

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

Table 1 Demographics ($N = 282$)

Variables	*N*	*%*
Gender		
Men	104	36.9
Women	178	63.1
Age		
Up to 22	144	51.1
Between 23 and 26	130	46.1
Above 26	8	2.8
Class enrollment		
Under-graduate	192	68.1
Graduate	88	31.2
Masters and above	2	0.7

3.2 Measurement

In this study, the variables were evaluated by using various scales already adopted by scholars to measure various constructs. The perceived ease of use was measured using a four-item scale developed by Davis *et al.* (1989). Similarly, the perceived usefulness was also assessed through the use of a four-item scale developed by Davis *et al.* (1989). To measure the subjective social norms, this study adopted a six-item scale developed by Hernandez *et al.* (2011). Consumer GBI was measured using a six-item scale developed by Venkatesh *et al.* (2012). Furthermore, the relational factors of trust and rapport were measured using a five-point Likert scale consisting of two items each, developed by Van Pinxteren *et al.* (2019). GACB was assessed using a six-item scale developed by Venkatesh *et al.* (2012). Finally, green OCB was measured through the use of a three-item scale developed by Cao *et al.* (2021).

4. Data analysis and results

4.1 Correlation analysis

As demonstrated in Table 2, the results indicated a significant positive correlation between perceived ease of use and GBI ($r = 0.13$, $p < 0.001$). Additionally, the results revealed a significant positive correlation between perceived usefulness and GBI ($r = 0.16$, $p < 0.001$). Furthermore, subjective social norms also had a positive correlation with GBI ($r = 0.5$,

Table 2 Descriptive statistics and correlation

Variable	*M*	*SD*	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*
PE-Use	26.75	4.06	–										
PE-Fullness	67.77	7.08	0.68***	–									
SS-Norms	48.52	6.10	0.61***	0.84**	–								
Trust	82.30	8.66	0.23***	0.68***	0.57**	–							
Rapport	58.89	7.20	0.12	0.45**	0.42**	0.54**	–						
GBI	65.90	9.15	0.26**	0.49**	0.38*	0.48**	0.60**	–					
GACB	41.30	4.95	0.14*	0.62**	0.58**	0.63**	0.69**	0.67***	–				
Green OCB	66.24	9.51	0.46**	0.59**	0.61*	0.41**	0.59**	0.56**	0.72**	–			
Gender	2.33	1.23	−0.15**	−0.06	−12	0.17*	0.31**	0.02	0.06	−08.	–		
Education	2.01	0.89	0.17*	0.32**	0.25**	0.35**	0.17**	0.20**	0.21**	0.14*	0.02	–	
Age	2.90	0.93	−17*	0.06	−0.02	0.31**	0.29**	0.06	0.17*	−0.04	0.53**	0.04	–

Notes: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$; PE-Use = Perceived ease of use; PU-Fullness = Perceived use fullness; SS-Norms = Subjective social norms; GBI = Green buying intention; GACB = Green actual consumption behavior; AG-OCB = Adolescent green organizational consumption behavior

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

$p < 0.01$). The results showed that the relational factors of trust ($r = 0.22$, $p < 0.01$) and rapport ($r = 0.44$, $p < 0.001$) were positively and significantly correlated with GBI, as indicated in Table 2.

4.2 Model measurement

Table 3 presents the mean and standard deviation of the demographic variables. To evaluate the model's compatibility, this study used a diverse range of fit indices, including goodness-of-fit measures, incremental fit measures, absolute fit measures and parsimonious fit indices. The selection of these measures is in accordance with the guidelines suggested by Khan *et al.* (2020a, 2020b). The validity of the survey questionnaire was established through the use of multiple tests, including maximum shared variance (MSV) and average shared variance (ASV). The item loadings were higher than the proposed standard value of 0.60 established by Fornell and Larcker (1981). The results in Table 3 of the reliability analysis showed that Cronbach's coefficients ranged from 0.773 to 0.954, which exceeded the 0.70 threshold value recommended by previous studies (Fornell and Larcker, 1981). Additionally, the results of the composite reliability analysis showed that coefficient of relative variation (CRV) ranged from 0.778 to 0.909, which was greater than the 0.70 standard predicted by Nunnally (1994) and Khan *et al.* (2020a, 2020b).

The results of the average variance extracted (AVE) analysis showed that the AVE ranged from 0.592 to 0.839, which was potentially higher than the 0.50 benchmark value suggested by Bagozzi *et al.* (1991) and slightly greater than the real MSV (Bagozzi *et al.*, 1991). Furthermore, the results indicated that the ASV was lower than the MSV and the AVE was higher than the threshold value.

4.3 Testing hypotheses

The primary aim of this study was to evaluate both the direct and indirect effects of the proposed variables, as delineated in the research hypotheses. To accomplish this objective, we used the PROCESS macro, supplemented by 5,000 bootstrap iterations. *H3a* posited that the association between perceived ease of use and GACB would be significantly mediated through GBI. The results of Model 6, as presented in Table 4, revealed that GBI significantly mediated the association between perceived ease of use and GACB ($\beta = 0.15$, 95% Confidence Interval [CI] = [−0.06, −0.13]). Additionally, the results showed that GBI significantly mediated the association between perceived usefulness and GACB ($\beta = 0.13$, 95% CI = [0.02, 0.37]).

However, the indirect effect between subjective social norms and GACB via GBI was found to be non-significant ($\beta = 0.01$, 95% CI = [0.03, −0.39]). On the other hand, the results indicated that the indirect effect of trust on GACB was significantly mediated through GBI

Table 3 Results of measurement analysis

Constructs	*FL*	*Cronbach's α*	*CRV*	*AVE*	*MSV*	*ASV*	*Skewness*	*Kurtosis*
Perceived ease of use	0.883–0.619	0.929	0.907	0.767	0.052	0.028	−0.991	1.919
Perceived usefulness	0.873–0.542	0.792	0.781	0.592	0.075	0.049	−0.902	1.892
Subjective social norms	0.749–0.812	0.851	0.863	0.675	0.475	0.214	−0.623	0.237
Trust	0.723–0.685	0.832	0.830	0.653	0.338	0.199	−0.802	1.782
Rapport	0.892–0.579	0.954	0.909	0.839	0.342	0.192	−0.809	1.729
GBI	0.686–0.797	0.773	0.778	0.690	0.053	0.029	−0.929	1.902
Green OCB	0.847–0.364	0.908	0.902	0.816	0.232	0.107	−0.867	1.355
GACB	0.782–0.847	0.877	0.899	0.794	0.492	0.289	−0.769	1.229

Notes: $p < 0.001$; GBI = Green buying intention; GACB = Green actual consumption behavior; OCB = Organizational consumption behavior; FL = Factor loadings; CRV = Composite reliability value; AVE = Average variance extracted; MSV = Maximum shared variance; ASV = Average shared variance

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

Table 4 Mediation analysis through process macro

Variables	*B*	*LLCI*	*ULCI*
Indirect effects of perceived ease of use on GACB			
Perceived ease of use GBI → GACB →	0.15*	−0.06	−0.13
Indirect effects of perceived fullness on GCCB			
Perceived usefulness → GBI GACB →	0.13**	0.02	0.37
Indirect effects of subjective social norm on GACB			
Subjective Social normal → GBI → GACB	0.01	0.03	−0.39
Indirect effects of Trust on GACB			
Trust GBI → GACB→	0.08*	0.19	0.62
Indirect effects of Repo on GACB			
Rapport GBI → GACB→	0.18**	0.07	0.28

Notes: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$; GBI = Green buying intention; GACB = Green actual consumption behavior; Green OCB = Organizational consumption behavior

($\beta = 0.08$, 95% CI = [0.19, 0.62]). Furthermore, the results revealed that the indirect effect of rapport on GACB was significantly mediated through GBI ($\beta = 0.18$, 95% CI = [0.07, 0.28]). These findings support *H3*, but only partially support *H4*, as GBI did not mediate the link between subjective social norms and GACB. As per *H5*, it was proposed that GBI would be positively associated with GACB. The results showed that GCBI was significantly positively linked with GACB ($r = 0.67$, $p < 0.01$). In conclusion, the results of this study provide evidence for the significance of GCBI in mediating the relationships between various variables and GACB and also highlight the positive association between GBI and GACB.

The results presented in Table 5 revealed that the interaction effect of young consumer GBI and green OCB on GACB was significant ($\beta = 0.22$, $t = 3.401$, $p < 0.001$). To further examine the moderating impact of adolescent green OCB, a plot of the interaction effect was produced. This finding highlights the significance of considering the interaction between GBI and adolescent green OCB in the prediction of GACB. The results suggest that the impact of GBI on GACB may vary based on the level of adolescent green OCB, highlighting the importance of considering both factors in understanding the association between GBI, adolescent green OCB and GACB.

Figure 2 provides an illustration of the interaction pattern between GBI and young consumer green OCB on young consumer GACB. Figure 2 depicts the association for two levels of adolescent green OCB: a higher level and a lower level. As predicted by *H6*, the association between GBI and GACB was found to be positive and the slope of the association increased as adolescent green OCB increased (high: $\beta = 0.79$, $p < 0.001$; low: $\beta = 0.31$, $p < 0.01$).

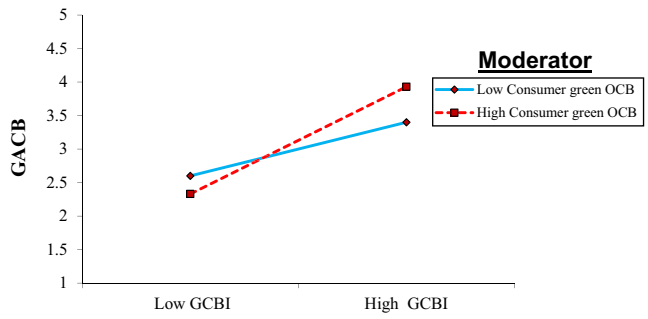
Table 5 Regression analysis and the moderating effect adolescent green OCB

Variables	B	SE	t	p	R^2/*Sig.*
Perceived ease of use on GACB	0.25	0.049	4.48	0.01	0.19**
Perceived use fullness on GACB	0.21	0.122	3.87	0.01	0.17**
Subjective Social normal on GACB	0.49	0.034	9.18	0.01	0.26**
Trust on GACB	0.22	0.029	4.32	0.01	0.19**
Rapport on GACB	0.12	0.098	2.34	0.01	0.24**
GBI on GACB	0.34	0.055	5.83	0.01	0.29**
Moderating Effect of CG-OCB on the relationship between GBI on GACB					
AG-OCB × GBI on GACB	0.22	0.061	3.401	0.01	0.28***

Notes: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$; GBI = Green buying intention; GACB = Green actual consumption behavior; AG-OCB = Adolescent green organizational consumption behavior

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

Figure 2 Interaction graph of moderating effect of green OCB



Source: Authors own work

These results provide evidence for the significance of considering the interaction between GBI and adolescent green OCB in the prediction of GACB. The positive association between GBI and GACB, combined with the increasing slope of the association as adolescent green OCB increases, supports H6 and highlights the importance of considering the interaction of these factors in understanding the association between GBI, adolescent green OCB and GACB.

5. Discussion

The current study proposed a total of six hypotheses to evaluate the research model, and the results of this investigation substantiated all of the proposed research hypotheses. Specifically, our findings supported the influence of technological and relational factors on young consumer GBI, thereby corroborating the *H1* and *H2*. These findings are consistent with prior research that has demonstrated the significant role of technological factors and trust as predictors of consumer intentions to make product purchases (Faqih, 2016). As a result, these findings contribute to the development of effective marketing strategies, highlighting the importance of leveraging technology and fostering trust as positive factors for shaping customer intentions.

Likewise, the hypotheses put forth in this study also demonstrated the significant mediating role of intention between technological factors and consumption behavior. These findings are consistent with prior research that has highlighted the significant influence of consumer intention in the context of e-commerce (Hidayanto *et al.*, 2017). Consequently, these results suggest that technology not only plays a crucial role in promoting green consumption behavior but also emphasizes the importance of consumer intention, particularly among young individuals in the digital age. Additionally, the findings of this study revealed the significant mediating role of young consumer GBI in the relationship between relational factors and GACB. This findings align with previous research conducted by Vijayasarathy (2004), which established a noteworthy influence of technology on consumer intention and purchasing behavior. Furthermore, these results underscored the significance of relationship factors such as trust and rapport in influencing consumer behavior outcomes, a notion that finds support in prior studies conducted by Bilgihan (2016), Fatima and Razzaque (2013).

Our study's findings also showed that young consumer GBI was a key mediator in the connection between technological and interpersonal elements and young consumer GACB in the online purchase mode. This result is in line with other study, which discovered that shopping online using contemporary digital gadgets had a strong mediating effect on

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

customers' purchase behavior. For instance, Steg and Vlek (2009) contend that social factors such as norms and values, as well as individual elements such as attitudes and beliefs, also have an impact on environmental behavior. They put out a theory known as the "multilevel perspective", which takes into account how societal and personal factors interact to shape behavior. Similar to this, recent studies have looked into how organizational factors, including green OCB, might help promote environmental behavior (Han and Kim, 2020; Ren *et al*., 2020).

Our study also found that green OCB has a significant moderating effect, and that increasing green OCB strengthens the association between young consumers' GBI and GACB when they shop online. This is in line with past work by our team that showed the significance of OCB in the relationship between customer intention and behavior in marketing contexts (Organ, 1997). Our research therefore broadens readers' comprehension by focusing on young people who use technology to make purchasing decisions. This is in line with other studies that claim GBI significantly influences the forecasting of a variety of behavioral outcomes, such as the acceptance and usage of technology (Davis, 1989).

Our study distinguishes itself by investigating the application of the TPB to online shopping within the context of sustainable consumption behavior among young individuals enrolled in diverse educational institutions. By examining the interplay between technological and relational factors in technology adoption, the TPB approach, as emphasized by Venkatesh *et al*. (2003), finds support in our findings. Furthermore, our study expands the scope of the TPB framework by incorporating novel aspects of actual consumption behavior in relation to young consumer intentions in the context of online shopping. Additionally, the present study employed a time lag approach, a methodology that has been widely used in previous research on customer intention and buying behaviors in the digital age. By using this approach, participants in our study had ample opportunities to record and revise their responses, thus enhancing the reliability and accuracy of the data collected.

5.1 Theoretical implications

This study offers several theoretical implications. The present study provides evidence that supports the use of the TPB in understanding young consumers' GACB. This study demonstrates that GBI, a key construct of TPB, is a significant predictor of young consumer GACB. This finding is consistent with previous research that has used TPB to predict environmental behavior among young consumers (Han *et al*., 2010; Kim-Soon *et al*., 2016). Lee *et al.* (2016) examined the factors influencing young consumers' intention to purchase green products and found that TPB was a useful framework for understanding young consumers' environmental behavior. Similarly, Han and Kim (2019) used TPB to investigate the determinants of young consumers' intention to purchase green products and found that TPB was a useful tool for predicting environmental behavior. This study extends previous research by examining the relationship between young consumer GBI and GACB, providing further evidence that TPB can be effectively used to understand young consumers' environmental behavior.

Second, this study highlights the importance of considering technological and relational factors in predicting young consumer GBI. The finding suggests that the TPB should be extended to incorporate these factors to better understand young consumers' environmental behavior. This is consistent with previous research that has suggested the need to extend TPB to incorporate additional factors beyond attitudes, subjective norms and perceived behavioral control (Bamberg and Möser, 2007; Kaiser and Craig, 2014). Bamberg and Möser (2007) argue that TPB should be extended to include additional factors such as values, habits and situational factors that may influence environmental behavior. Similarly, Kaiser and Wilson (2004) suggest that TPB should be extended to include factors such as environmental concern, knowledge and access to environmental

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

information. This study contributes to the literature by demonstrating the importance of incorporating technological and relational factors into the TPB framework to better understand young consumers' environmental behavior. Finally, this study demonstrates the moderating effect of young green OCB on the association between young consumer GBI and young consumer GACB. This finding suggests that the TPB can be enhanced by considering the role of green OCB in predicting environmental behavior. This is consistent with previous research that has highlighted the importance of considering the social context in shaping environmental behavior (Steg and Vlek, 2009). This study contributes to the literature by demonstrating the moderating effect of young green OCB on the association between young consumer GBI and young consumer GACB. This finding suggests that incorporating the role of organizational factors, such as green OCB, into the TPB framework can enhance its predictive power for environmental behavior among young consumers.

5.2 Managerial implications

The present study highlights the importance of technological and relational factors in influencing young consumers' GACB and GBI. This study provides valuable insights for managers looking to enhance young consumers' GACB in the context of online shopping.

First, the results of this study emphasize the significance of technological factors, such as perceived ease of use and perceived usefulness, in shaping young consumers' GBI. Managers can leverage this knowledge by investing in improving the technology infrastructure of their online platforms and making sure that these platforms is user-friendly and easy to use. Furthermore, this study underscores the significance of perceived usefulness by emphasizing the requirement for managers to furnish young consumers with informative and instructional resources that impart the environmental advantages of green products. This underscores the need for managers to prioritize the provision of educational materials that help to increase awareness of the importance of green products and their positive impact on the environment. By doing so, they can play a crucial role in promoting the adoption of eco-friendly products among young consumers and contributing to a more sustainable future.

Second, this study also highlights the importance of relational factors, such as trust and rapport, in shaping young consumers' GBI. The findings of this study can be used by managers in building trust and establishing a positive association with young consumers. This can be achieved through the implementation of transparent communication and the adoption of sustainable business practices. By fostering transparency and demonstrating a commitment to sustainability, managers can not only build trust with young consumers but also establish a positive reputation for their business. This, in turn, can lead to increased brand loyalty and improved market performance in the long term. It is thus imperative for managers to invest in building trust and rapport with young consumers through sustainable and transparent business practices.

Third, this study highlights the mediating role of young consumers' GBI in the association between technological and relational factors and young consumers' GACB. The insights gained from this study can be used by managers in the design of targeted marketing campaigns that aim to influence the young consumers' GBI toward eco-friendly products. By focusing their efforts on shaping young consumers' GBI, managers can effectively promote the adoption of green products and contribute to the development of a more sustainable future. To achieve this, managers should consider the unique perspectives, preferences and motivations of young consumers when designing their marketing strategies. This will enable them to effectively communicate the benefits of green products and persuade young consumers to adopt eco-friendly alternatives. It is therefore imperative for managers to use the information obtained from this study in the design of targeted marketing campaigns that aim to shape young consumers' GBI toward green products.

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

Finally, this study highlights the moderating role of adolescent green OCB in the association between young consumers' GBI and GACB. Managers can leverage this knowledge by investing in programs that encourage young consumers to engage in pro-environmental behaviors, such as recycling and reducing energy usage, and promoting the environmental benefits of these behaviors. The present study provides valuable insights for managers looking to enhance young consumers' GACB in the context of online shopping. By leveraging the findings of this study, managers can design effective marketing strategies that target young consumers and increase their engagement with green products.

5.3 Limitations and future research directions

The present study has several limitations that should be considered when interpreting the results. First, the sample used in this study was limited to young consumers who engaged in online shopping; therefore the findings may not be generalizable to other populations or contexts. Second, this study used a cross-sectional design, which limits the ability to establish causality between the variables of interest. Third, this study relied on self-reported measures of GACB, GBI and green OCB, which may be subject to biases such as social desirability bias and memory recall bias.

Additionally, the use of the TPB framework in this study was limited to a single theoretical model and did not take into consideration other relevant theories and models in the field of green consumer behavior. Further research could consider incorporating other theories and models to expand the understanding of the factors that influence young consumers' green buying behavior.

It is also important to note that while this study found evidence of a significant association between GBI and GACB, as well as the mediating and moderating effects of GBI and green OCB, respectively, further research is needed to understand the nature and mechanism of these relationships. Further studies could also explore the impact of other factors such as cultural, social and psychological factors on young consumers' green buying behavior.

References

Ahmed, N., Li, C., Khan, A., Qalati, S.A., Naz, S. and Rana, F. (2021), "Purchase intention toward organic food among young consumers using theory of planned behavior: role of environmental concerns and environmental awareness", *Journal of Environmental Planning and Management*, Vol. 64 No. 5, pp. 796-822.

Ajzen, I. (1985), "From intentions to actions: a theory of planned behavior", in Kuhi, J. and Beckmann, J. (Eds), *Action-Control: From Cognition to Behavior*, Springer, pp. 11-39.

Ajzen, I. (1991), "The theory of planned behavior", *Organizational Behavior and Human Decision Processes*, Vol. 50 No. 2, pp. 179-211.

Ajzen, I. (2005), *Attitudes, Personality, and Behavior*, 2nd ed, McGraw-hill Education, UK.

Ajzen, I. (2011), "The theory of planned behaviour: reactions and reflections", *Psychology & Health*, Vol. 26 No. 9, pp. 1113-1127.

Ali, A., Hameed, A., Moin, M.F. and Khan, N.A. (2022), "Exploring factors affecting mobile-banking app adoption: a perspective from adaptive structuration theory", *Aslib Journal of Information Management*. doi: 10.1108/AJIM-08-2021-0216

Anderson, J.C. and Narus, J.A. (2004), "A model of distributor firm and manufacturer firm working partnerships", *Journal of Marketing*, Vol. 68 No. 1, pp. 1-13.

Babalola, M.T., Greenbaum, R.L., Amarnani, R.K., Shoss, M.K., Deng, Y., Garba, O.A. and Guo, L. (2020), "A business frame bottom-line, perspective on why perceptions of top management's mentality result in employees' good and bad behaviors", *Personnel Psychology*, Vol. 73 No. 1, pp. 19-41.

Bagozzi, R.P., Yi, Y. and Phillips, L.W. (1991), "Assessing construct validity in organizational research", *Administrative Science Quarterly*, Vol. 36 No. 3, pp. 421-458.

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

Bamberg, S. and Möser, G. (2007), "Twenty years after Hines, Hungerford, and Tomera: a new meta-analysis of psycho-social determinants of pro-environmental behaviour", *Journal of Environmental Psychology*, Vol. 27 No. 1, pp. 14-25.

Barbarossa, C. and Pastore, A. (2015), "Why environmentally conscious consumers do not purchase green products", *Qualitative Market Research: An International Journal*, Vol. 18 No. 2, pp. 188-209.

Bilgihan, A. (2016), "Gen Y customer loyalty in online shopping: an integrated model of trust, user experience and branding", *Computers in Human Behavior*, Vol. 61, pp. 103-113, doi: 10.1016/j.chb.2016.03.014.

Brislin, R.W. (1980), "Translation and content analysis of oral and written materials", *Methodology*, pp. 389-444.

Cao, D., Meadows, M., Wong, D. and Xia, S. (2021), "Understanding consumers' social media engagement behaviour: an examination of the moderation effect of social media context", *Journal of Business Research*, Vol. 122, pp. 835-846.

Chang, C.C. (2011), "Understanding the antecedents of green purchase behavior: an exploratory study in Taiwan", *Journal of Business Ethics*, Vol. 101 No. 4, pp. 579-593.

Chen, Y.C. and Chang, H.H. (2014), "The effect of subjective norm on environmental behavior: the case of electronic waste recycling", *Journal of Business Ethics*, Vol. 121 No. 1, pp. 121-132.

Chen, Y.S. and Chang, C.H. (2012), "Enhance green purchase intentions: the roles of green perceived value, green perceived risk, and green trust", *Management Decision*, Vol. 50 No. 3, pp. 502-520.

Chen, S.C. and Dhillon, G.S. (2003), "Interpreting dimensions of consumer trust in e-commerce", *Information Technology and Management*, Vol. 4 Nos 2/3, pp. 303-318.

Davis, F.D. (1985), *A Technology Acceptance Model for Empirically Testing New End-User Information Systems: Theory and Results*, Massachusetts Institute of Technology.

Davis, F.D. (1989), "Perceived usefulness, perceived ease of use, and user acceptance of information technology", *MIS Quarterly*, Vol. 13 No. 3, pp. 319-340.

Davis, F.D., Bagozzi, R.P. and Warshaw, P.R. (1989), "User acceptance of computer technology: a comparison of two theoretical models", *Management Science*, Vol. 35 No. 8, pp. 982-1003.

Fan, A., Wu, L., Miao, L. and Mattila, A.S. (2020), "When does technology anthropomorphism help alleviate customer dissatisfaction after a service failure?–The moderating role of consumer technology self-efficacy and interdependent self-construal", *Journal of Hospitality Marketing & Management*, Vol. 29 No. 3, pp. 269-290.

Faqih, K.M.S. (2016), "Journal of retailing and consumer services an empirical analysis of factors predicting the behavioral intention to adopt internet shopping technology among non-shoppers in a developing country contex: does gender matter?", *Journal of Retailing and Consumer Services*, Vol. 30, pp. 140-164, doi: 10.1016/j.jretconser.2016.01.016.

Fatima, J.K. and Razzaque, M.A. (2013), "Roles of customer involvement in rapport and satisfaction", *Asia Pacific Journal of Marketing and Logistics*, Vol. 25 No. 3, pp. 452-471.

Fornell, C. and Larcker, D.F. (1981), "Structural equation models with unobservable variables and measurement error: algebra and statistics", *Journal of Marketing Research*, Vol. 18 No. 3, pp. 382-388.

Gao, B., Li, Z. and Yan, J. (2022), "The influence of social commerce on eco-friendly consumer behavior: technological and social roles", *Journal of Consumer Behaviour*, Vol. 21 No. 4, pp. 653-672.

Gremler, D.D. and Gwinner, K.P. (2000), "Customer-employee rapport in service relationships", *Journal of Service Research*, Vol. 3 No. 1, pp. 82-104.

Guath, M., Stikvoort, B. and Juslin, P. (2022), "Nudging for eco-friendly online shopping–attraction effect curbs price sensitivity", *Journal of Environmental Psychology*, Vol. 81, p. 101821.

Han, H. and Kim, Y. (2019), "An investigation of green hotel customers' decision formation: developing an extended model of the theory of planned behavior", *International Journal of Hospitality Management*, Vol. 29 No. 4, pp. 659-668.

Han, H. and Kim, Y. (2020), "Chatbot acceptance model using the self-determination theory", *Computers in Human Behavior*, Vol. 103, pp. 213-223.

Han, H., Hsu, L.T.J. and Sheu, C. (2010), "Application of the theory of planned behavior to green hotel choice: testing the effect of environmental friendly activities", *Tourism Management*, Vol. 31 No. 3, pp. 325-334.

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

Hernandez, B., Montaner, T., Sese, F.J. and Urquizu, P. (2011), "The role of social motivations in e-learning: how do they affect usage and success of ICT interactive tools?", *Computers in Human Behavior*, Vol. 27 No. 6, pp. 2224-2232.

Hidayanto, A.N., Ovirza, M., Anggia, P., Budi, N.F.A. and Phusavat, K. (2017), "The roles of electronic word of mouth and information searching in the promotion of a new e-commerce strategy: a case of online group buying in Indonesia", *Journal of Theoretical and Applied Electronic Commerce Research*, Vol. 12 No. 3, pp. 69-85.

Hollebeek, L.D. and Macky, K. (2019), "Digital content marketing's role in fostering consumer engagement, trust, and value: framework, fundamental propositions, and implications", *Journal of Interactive Marketing*, Vol. 45 No. 1, pp. 27-41.

Hu, Y. and Lu, L. (2021), "The moderating effect of green organizational citizenship behavior on the relationship between perceived ease of use and green intention in online shopping", *Journal of Business Research*, Vol. 123, pp. 324-331.

Huang, J., Wang, H., Chen, X. and Dai, J. (2021), "Exploring the role of self-efficacy in predicting gig work adoption", *International Journal of Human Resource Management*, Vol. 32 No. 3, pp. 680-699.

Kaiser, R.B. and Craig, S.B. (2014), "Destructive leadership in and of organizations", in Day, D.V. (Ed.), *The Oxford Handbook of Leadership and Organizations*, Oxford University Press, pp, pp. 260-284.

Kaiser, F.G. and Wilson, M. (2004), "Goal-directed conservation behavior: the specific composition of a general performance", *Personality and Individual Differences*, Vol. 36 No. 7, pp. 1531-1544.

Khan, A.N. and Khan, N.A. (2021), "The nexuses between transformational leadership and employee green organisational citizenship behaviour: role of environmental attitude and green dedication", *Business Strategy and the Environment*, Vol. 31 No. 3, pp. 921-933.

Khan, S.K., Ali, N., Khan, N.A., Ammara, U. and Anjum, N. (2020a), "Understanding multiscreening phenomenon for online shopping through perspective of self-regulation and dual process theory: case of Chinese young generation", *Electronic Commerce Research and Applications*, Vol. 42, p. 100988.

Khan, S.K., Khan, N.A., Egorova, O., Zia, N. and Akhtar, M. (2023), "Young consumers' usability preferences for mobile and computer screens for online shopping activities", *Human Behavior and Emerging Technologies*, Vol. 2023.

Khan, A.N., Moin, M.F., Khan, N.A. and Zhang, C. (2022), "A multistudy analysis of abusive supervision and social network service addiction on employee's job engagement and innovative work behaviour", *Creativity and Innovation Management*, Vol. 31 No. 1, pp. 77-92, doi: 10.1111/CAIM.12481.

Khan, S.A.R., Yu, Z., Belhadi, A. and Mardani, A. (2020b), "Investigating the effects of renewable energy on international trade and environmental quality", *Journal of Environmental Management*, Vol. 272, p. 111089.

Kim, H.J. and Ko, E. (2010), "The effect of perceived social norms on technology acceptance: an extension of the theory of planned behavior", *International Journal of Human-Computer Interaction*, Vol. 26 No. 2, pp. 107-124.

Kim, Y. and Lee, J. (2020), "The moderating effect of green organizational citizenship behavior on the relationship between perceived usefulness and green intention in online shopping", *Sustainability*, Vol. 12 No. 13, p. 4976.

Kim, B., Lee, G., Murrmann, S.K. and George, T.R. (2012), "Motivational effects of empowerment on employees' organizational commitment: a mediating role of management trustworthiness", *Cornell Hospitality Quarterly*, Vol. 53 No. 1, pp. 10-19.

Kim-Soon, N.G.A.R.A., Ahmad, A.R. and Ibrahim, N.N. (2016), "Theory of planned behavior: undergraduates' entrepreneurial motivation and entrepreneurship career intention at a public university", *Journal of Entrepreneurship: Research & Practice*, pp. 1-14.

Kim, T. and Yun, S. (2019), "How will changes toward pro-environmental behavior play in customers' perceived value of environmental concerns at coffee shops", *Sustainability*, Vol. 11 No. 14, p. 3816.

Kollmuss, A. and Agyeman, J. (2002), "Mind the gap: why do people act environmentally and what are the barriers to pro-environmental behavior?", *Environmental Education Research*, Vol. 8 No. 3, pp. 239-260.

Kwon, H.J. and Wen, Y. (2017), "Understanding technology adoption and use behaviors of young people: the role of attitudes towards technology", *Telematics and Informatics*, Vol. 34 No. 2, pp. 259-268.

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

Lee, J., Kim, S. and Ham, C.D. (2016), "A double-edged sword? Predicting consumers' attitudes toward and sharing intention of native advertising on social media", *American Behavioral Scientist*, Vol. 60 No. 12, pp. 1425-1441.

Lee, Y.K., Lee, J.Y. and Choi, J.N. (2012), "An extended technology acceptance model for green IT adoption", *Decision Support Systems*, Vol. 53 No. 1, pp. 218-228.

Liang, J. and Li, Y. (2017), "Green intention and green consumption behavior: a study of young consumers in China", *Journal of Cleaner Production*, Vol. 142, pp. 3344-3355.

Lin, C.H. and Hsiao, C.Y. (2020), "Green intention, environmental attitudes, and green consumption behavior among college students in Taiwan", *Sustainability*, Vol. 12 No. 2, p. 545.

Lin, J.C. and Lu, H.P. (2011), "Understanding young consumers' green purchase behavior: an examination of the roles of environmental concern, ecological responsibility, and personal values", *Journal of Business Ethics*, Vol. 102 No. 1, pp. 133-148.

Liu, Y. and Chen, X. (2022), "The moderating effect of green organizational citizenship behavior on the relationship between environmental concern and green consumption behavior in online shopping", *Sustainability*, Vol. 14 No. 8, p. 3066.

Ng, P.M.L. and Cheung, C.T.Y. (2022), "Why do young people do things for the environment? The effect of perceived values on pro-environmental behaviour", *Young Consumers*, Vol. 23 No. 4, pp. 539-554.

Nunnally, J.C. (1994), *Psychometric Theory 3E*, Tata McGraw-hill education.

Organ, D.W. (1988), *Organizational Citizenship Behavior: The Good Soldier Syndrome*, Lexington Books/DC Heath and Com.

Organ, D.W. (1997), "Organizational citizenship behavior: it's construct cleanup time", *Human Performance*, Vol. 10 No. 2, pp. 85-97.

Park, J., Lee, Y. and Lee, J. (2014), "The effect of perceived usefulness, perceived ease of use, and trust on green buying behavior: an empirical study", *Sustainability*, Vol. 6 No. 9, pp. 6869-6889.

Qi, X. and Ploeger, A. (2019), "Explaining consumers' intentions towards purchasing green food in Qingdao, China: the amendment and extension of the theory of planned behavior", *Appetite*, Vol. 133, pp. 414-422.

Qi, X., Tian, X. and Ploeger, A. (2021), "Exploring Chinese consumers' online purchase intentions toward certified food products during the COVID-19 pandemic", *Foods*, Vol. 10 No. 11, p. 2729.

Rana, J. and Paul, J. (2017), "Consumer behavior and purchase intention for organic food: a review and research agenda", *Journal of Retailing and Consumer Services*, Vol. 38, pp. 157-165.

Ren, S., Chadee, D. and Presbitero, A. (2020), "Influence of informal relationships on expatriate career performance in China: the moderating role of cultural intelligence", *Management and Organization Review*, Vol. 16 No. 3, pp. 569-593.

Steg, L. and Vlek, C. (2009), "Encouraging pro-environmental behaviour: an integrative review and research agenda", *Journal of Environmental Psychology*, Vol. 29 No. 3, pp. 309-317.

Turchi, B. and Steg, L. (2009), "Green consumer behavior: a review of the literature", *Ecological Economics*, Vol. 68 No. 3, pp. 658-668.

Van Pinxteren, M.M., Wetzels, R.W., Rüger, J., Pluymaekers, M. and Wetzels, M. (2019), "Trust in humanoid robots: implications for services marketing", *Journal of Services Marketing*, Vol. 33 No. 4, pp. 507-518.

Venkatesh, V. and Davis, F.D. (2000), "A theoretical extension of the technology acceptance model: four longitudinal field studies", *Management Science*, Vol. 46 No. 2, pp. 186-204.

Venkatesh, V., Morris, M.G., Davis, G.B. and Davis, F.D. (2003), "User acceptance of information technology: toward a unified view", *MIS Quarterly*, Vol. 27 No. 3, pp. 425-478.

Venkatesh, V., Thong, J.Y. and Xu, X. (2012), "Consumer acceptance and use of information technology: extending the unified theory of acceptance and use of technology", *MIS Quarterly*, Vol. 36 No. 1, pp. 157-178.

Vijayasarathy, L.R. (2004), "Predicting consumer intentions to use on-line shopping: the case for an augmented technology acceptance model", *Information & Management*, Vol. 41 No. 6, pp. 747-762.

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS

Wang, C.H. (2019), "How organizational green culture influences green performance and competitive advantage: the mediating role of green innovation", *Journal of Manufacturing Technology Management*, Vol. 30 No. 4, pp. 666-683.

Wang, D. and Chen, H. (2009), "The impact of vendor trust and rapport on consumers' green purchase behavior", *Journal of Business Ethics*, Vol. 88 No. 3, pp. 519-532.

Wang, S., Chen, Z., Xiao, Y. and Lin, C. (2021), "Consumer privacy protection with the growth of AI-empowered online shopping based on the evolutionary game model", *Frontiers in Public Health*, Vol. 9, p. 705777.

Wang, J. and Kim, T.Y. (2013), "Proactive socialization behavior in China: the mediating role of perceived insider status and the moderating role of supervisors' traditionality", *Journal of Organizational Behavior*, Vol. 34 No. 3, pp. 389-406.

Wang, D., Liang, T.P., Li, Y. and Liang, D. (2013), "Understanding the factors influencing the adoption of green IT in organizations: a technology acceptance perspective", *Computers in Human Behavior*, Vol. 29 No. 6, pp. 2244-2257.

Wei, Y. and Lee, D. (2014), "The role of trust and green intention in green purchasing behavior", *Journal of Business Ethics*, Vol. 121 No. 4, pp. 557-564.

White, M.D. and Fisher, C. (2008), "Assessing our knowledge of identity theft: the challenges to effective prevention and control efforts", *Criminal Justice Policy Review*, Vol. 19 No. 1, pp. 3-24.

Xu, X. and Zhang, J. (2023), "The moderating effect of green organizational citizenship behavior on the relationship between eco-friendly attitudes and green consumption behavior in online shopping", *Journal of Business Ethics*, Vol. 176 No. 2, pp. 333-344.

Yuan, S., Liu, Y., Yao, R. and Liu, J. (2016), "An investigation of users' continuance intention towards mobile banking in China", *Information Development*, Vol. 32 No. 1, pp. 20-34.

Zhang, J. and Wang, Y. (2022), "The moderating effect of green organizational citizenship behavior on the relationship between perceived risk and green intention in online shopping", *Journal of Cleaner Production*, Vol. 282, p. 124285.

Zhang, Y., Wang, J., Xue, Y. and Yang, J. (2018), "Impact of environmental regulations on green technological innovative behavior: an empirical study in China", *Journal of Cleaner Production*, Vol. 188, pp. 763-773.

Zhao, H., Wang, X. and Jiang, L. (2021), "To purchase or to remove? Online shopping cart warning pop-up messages can polarize liking and purchase intention", *Journal of Business Research*, Vol. 132, pp. 813-836.

Corresponding author

Naseer Abbas Khan can be contacted at: naseer@mail.ustc.edu.cn

For instructions on how to order reprints of this article, please visit our website:
www.emeraldgrouppublishing.com/licensing/reprints.htm
Or contact us for further details: permissions@emeraldinsight.com

Downloaded from http://www.emerald.com/yc/article-pdf/24/6/686/2419539/yc-09-2022-1610.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCS